

FIRST TERM EXAMINATION 2081

Class: XII 'Science'

SET-B

Time: 25 min

Subject: Physics [0031]

Candidates are required to give their answer in their own words as far as practicable. The figure in the margin indicated full marks:

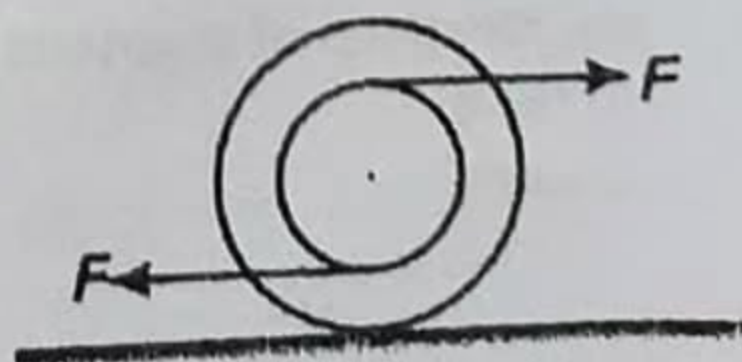
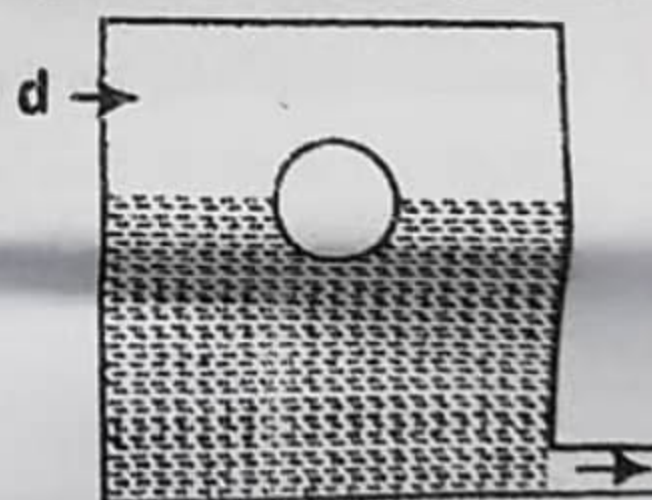
Attempt all the questions.

GROUP-A [11*2=22]

KMC Note

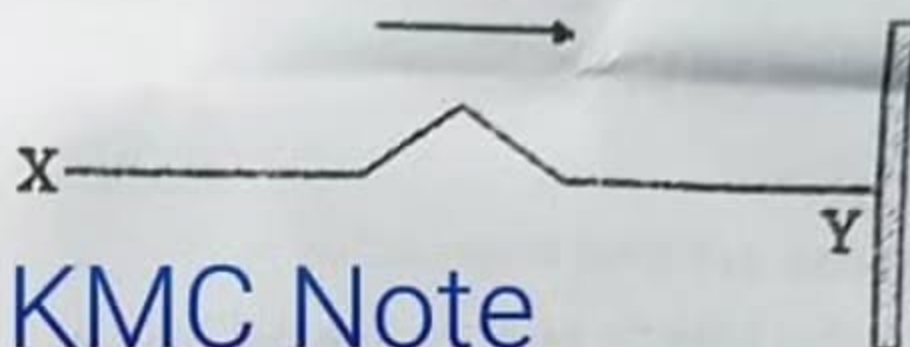
Choose the best alternative

1. The kinetic energy of a particle executing SHM changes from maximum to minimum in 6 s. Then the time period of SHM is :
A. 6 s B. 12 s C. 18 s D. 24 s
2. A vessel contains water over mercury (density = 13.6 gm/cm^3). A uniform sphere floats with half its volume immersed in mercury and the other half in oil. The specific gravity of the material of sphere is
A. 3.3 B. 6.4 C. 7.3 D. 12.8
3. A metal shell loses one third of its weight in water. Relative density of metal is 6, what percentage of its volume is hollow?
A. 50% B. 30% C. 20% D. 60%
4. A tank filled with water has two taps to exhaust and pour. A hollow spherical ball is half submerged in water. Through one tap, water is taken out and through another tap, a liquid of density double the density of water is poured in tank such that volume of liquid in tank remains constant. Sphere will
A. go down B. go up
C. maintain same height D. sink to bottom
5. A spool is pulled horizontally by two equal and opposite forces as shown in the figure. Which of the following statements is correct?
A. The centre of mass moves towards left
B. The centre of mass moves towards right
C. The centre of mass remains stationary
D. The net torque about the centre of mass of the spool is zero.



6. The phase difference between two points separated by 0.8 m in a wave of frequency 120 Hz is 0.5π . The velocity of wave will be
 A. 720 m/s B. 384 m/s C. 256 m/s D. 144 m/s
7. The end correction for the vibration of air column in a tube of circular cross section will be more if the tube is
 a. increased in length b. decreased in length
 c. increased in diameter d. decreased in diameter

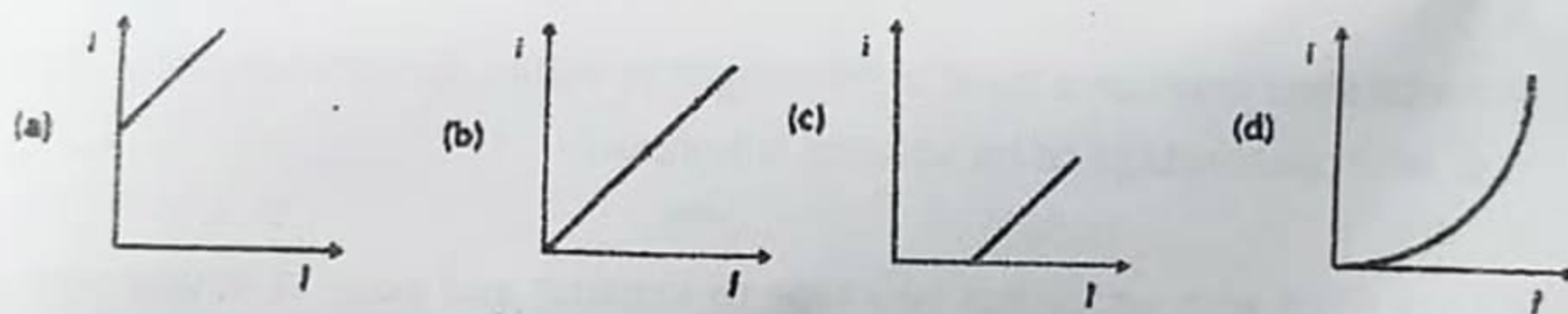
8. The diagram below shows a pulse travelling along a rope from X to Y. The end Y of the rope is tied to a fixed support.



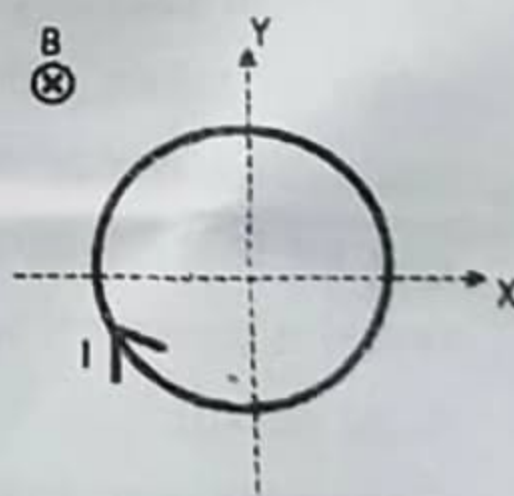
When the pulse reaches end Y it will

KMC Note

- A. disappear.
 B. cause the end of the rope at Y to oscillate up and down.
 C. be reflected and be inverted.
 D. be reflected and not be inverted.
9. The graph between intensity (I) of light falling on a metallic plate with the current (i) generated is



10. A conducting loop carrying a current I is placed in a uniform magnetic field pointing into the plane of the paper as shown. The loop will have a tendency to



- A. contract
 B. expand
 C. move towards +ve x-axis
 D. move towards -ve x-axis
11. In a high tension wire electric current runs from East to West. Then the direction of magnetic field at points below the wire is towards
 A. North B. south
 C. east D. west

FIRST TERM EXAMINATION 2081

Class: XII 'Science'

SET-A

Time: 25 min

Subject: Physics [0031]

Candidates are required to give their answer in their own words as far as practicable. The figure in the margin indicated full marks:

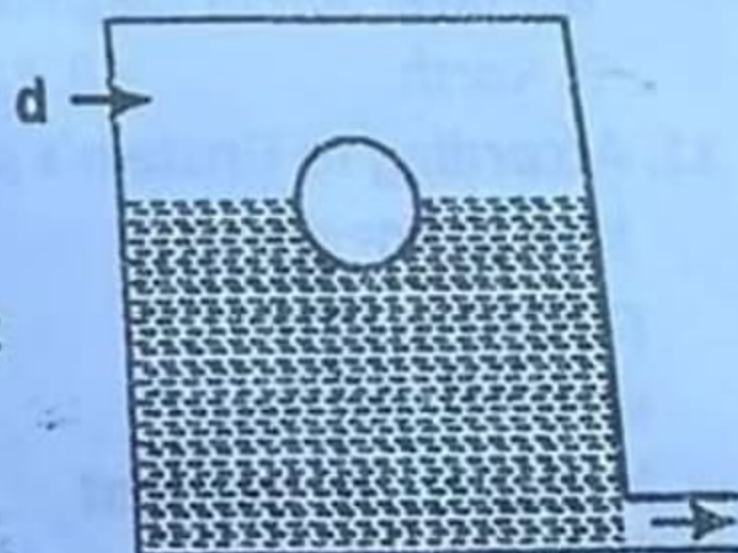
Attempt all the questions.

KMC Note

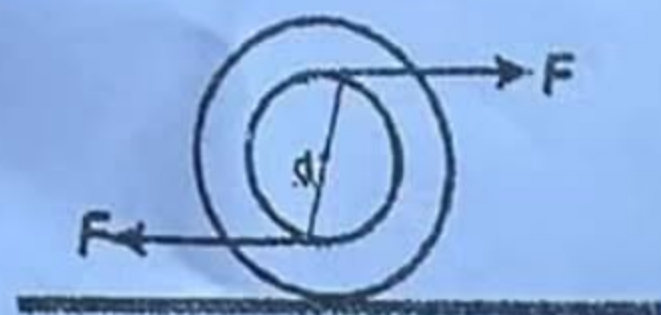
GROUP-A [11*1=11]

Choose the best alternative.

1. The potential energy of a particle executing SHM changes from maximum to minimum in 5 s. Then the time period of SHM is:
A. 5 s B. 10 s C. 15 s D. 20 s
2. A vessel contains oil (density = 0.8 gm/cm^3) over mercury (density = 13.6 gm/cm^3). A uniform sphere floats with half its volume immersed in mercury and the other half in oil. The specific gravity of the material of sphere is
A. 3.3 B. 6.4 C. 7.2 D. 12.8
3. A metal shell loses half its weight in water. Relative density of metal is 5, what fraction of its volume is hollow?
A. $3/5$ B. $2/5$ C. $1/5$ D. $4/5$
4. A tank filled with water has two taps to exhaust and pour. A hollow spherical ball is half submerged in water. Through one tap, water is taken out and through another tap, a liquid of density half the density of water is poured in tank such that volume of liquid in tank remains constant. Sphere will



- A. move down B. move up C. maintain same height D. sink to bottom
5. A spool is pulled horizontally by two equal and opposite forces (F) as shown in the figure. Magnitude of the net torque produced on the spool is (d is distance between the forces)



A. Zero

B. dF

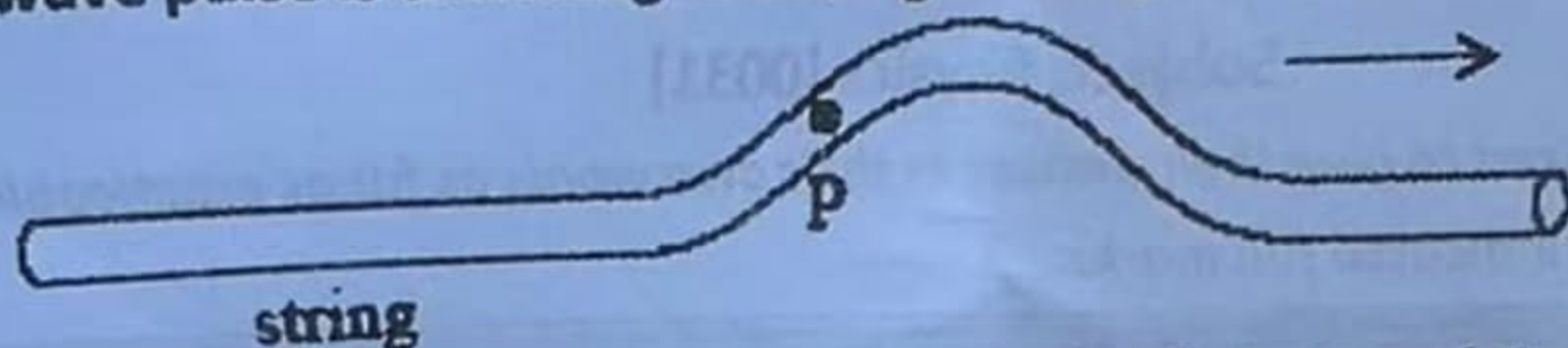
C. $2dF$

D. $4dF$

6. The distance between two points differing in phase by 60° on a wave having wave velocity 360 m/s and frequency 500 Hz is

A. 0.36 m B. 0.18 m C. 0.48 m D. 0.12 m

7. A wave pulse is travelling to the right along a string.



Which of the following best represents the direction of the velocity of the point P?

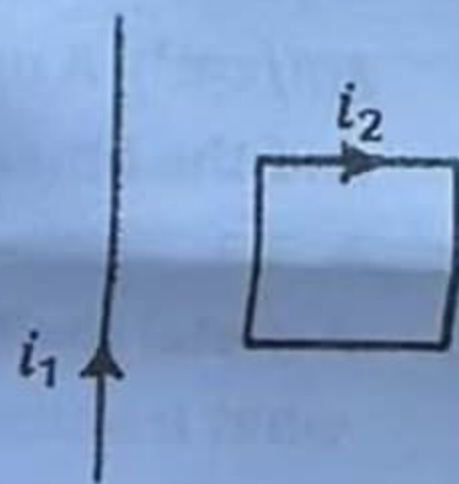
A. $\uparrow$ B. $\downarrow$ C. $\rightarrow$ D. $\leftarrow$

8. The end correction for the vibration of air column in a tube of circular cross section will be less if the tube is

A. made shorter B. made longer
C. made thinner D. made wider

9. Consider the situation shown in figure. The straight wire is fixed but the loop can move under magnetic force. The loop will

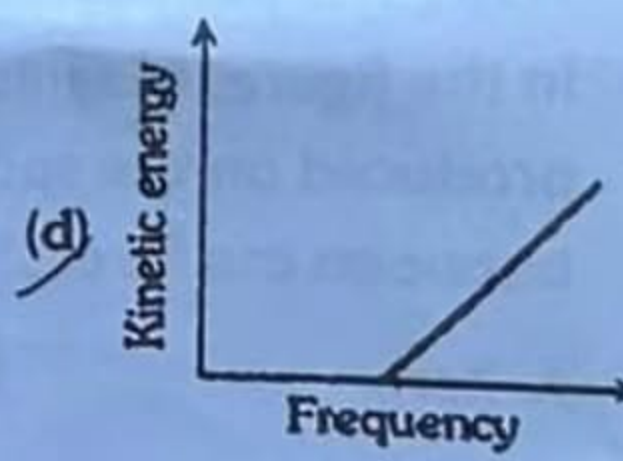
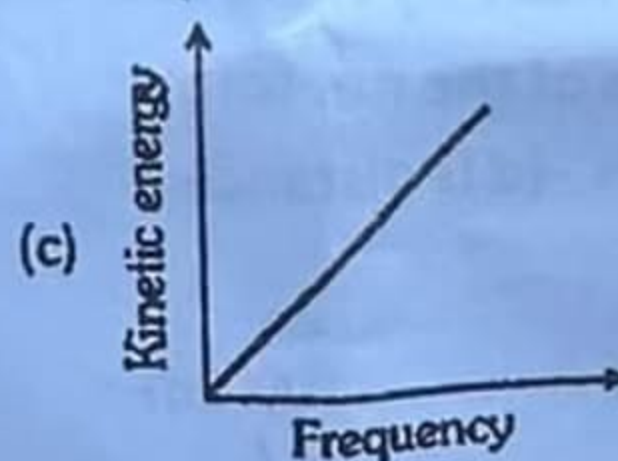
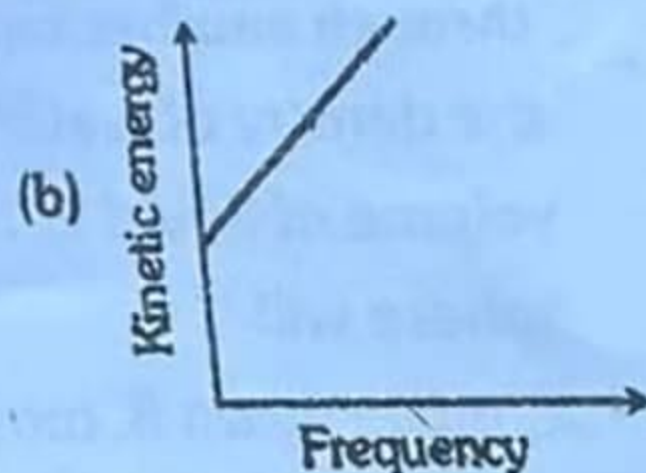
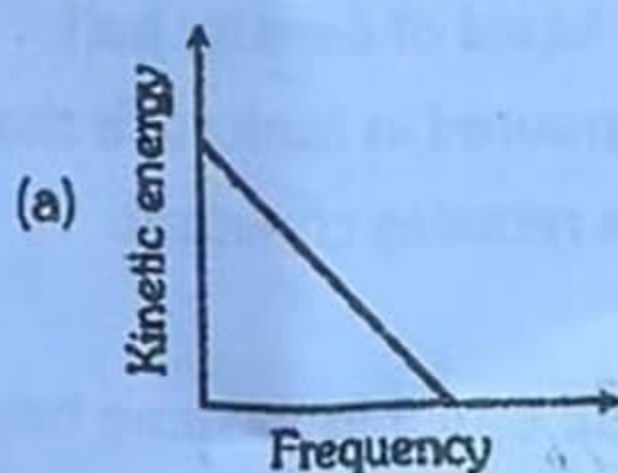
A. remains stationary
B. move towards the wire
C. move away from the wire
D. rotates about the wire



10. In a high tension wire electric current runs from East to West. Then the direction of magnetic field at points above the wire is towards

A. North B. south C. east D. west

11. According to Einstein's photoelectric equation, the graph between the kinetic energy of photoelectrons ejected and the frequency of incident radiation is



FIRST TERM EXAMINATION 2081

Class: XII 'Science'

F.M.: 75

Time: 3 hrs.

Subject: Physics [1021]

P.M.: 30

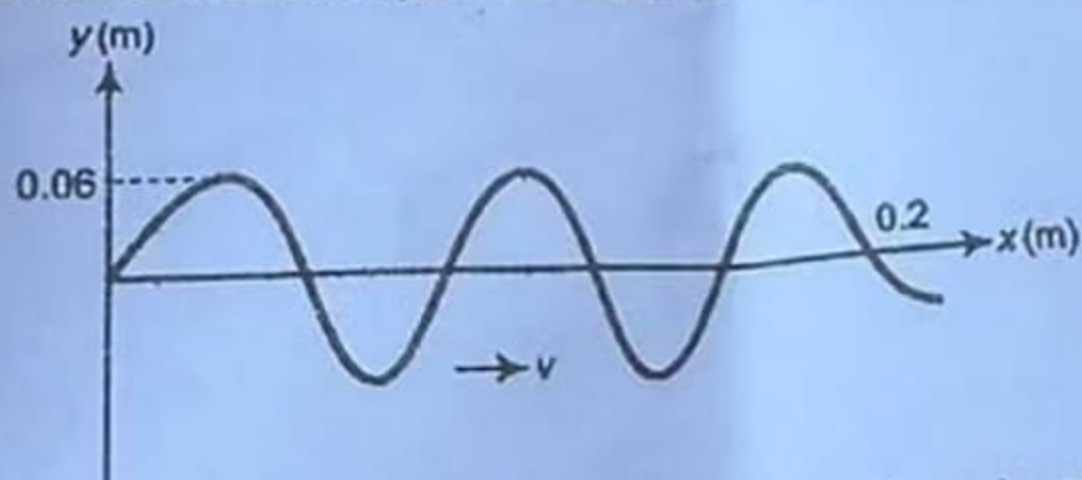
Candidates are required to give their answer in their own words as far as practicable. The figure in the margin indicated full marks:

Attempt all the questions.

GROUP-B [8×5=40]

12. a. Show that motion of a simple pendulum is simple harmonic motion and its time period is independent of mass of the bob. [3]
b. A particle executes simple harmonic motion about the point $x=0$. At time $t=0$, it has displacement $x=2\text{cm}$ and zero velocity. If the frequency of motion is 0.25 s^{-1} , find the displacement at $t=3\text{ s}$. [2]

13. a. Starting with the definition of stationary wave, prove that the distance between any two consecutive nodes or antinodes in a stationary wave is $\lambda/2$. [3]
b. For the wave shown in figure, write the equation of this wave if its position is shown at $t=0$. Speed of wave is $v=300\text{ m/s}$. [2]

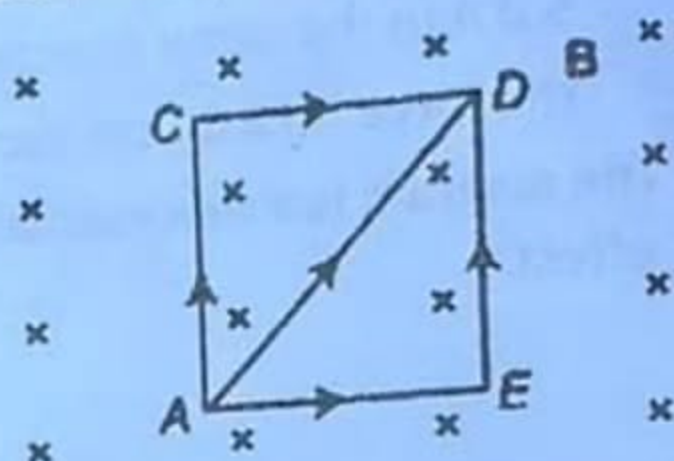


14. a. What are the assumptions made by Laplace to correct Newton's formula for velocity of sound in air? [2]
b. At what temperature will the speed of sound in hydrogen be the same as in oxygen at 100°C ? Molar masses of oxygen and hydrogen are in the ratio 1:16. [3]
15. a. Show that, in open organ pipe, both odd and even harmonics can be produced. [3]
b. A tube 1.0 m long is closed at one end. A stretched wire is placed near the open end. The wire is 0.3m long and has a mass of 0.01 kg. It is held fixed at both ends and vibrates in its fundamental mode. It sets the air column in

the tube into vibration at its fundamental frequency by resonance. Find the tension in the wire. (Speed of sound in air = 330 m/s.) [2]

16. a. What is the Hall effect? Find the expression for the Hall voltage. [3]

b. A square of side 2.0 m is placed in a uniform magnetic field $B = 2.0 \text{ T}$ in a direction perpendicular to the plane of the square inwards. Equal current $i = 3.0 \text{ A}$ is flowing in the directions shown in figure. Find the magnetic force on the loop. [2]



17. a. Explain the construction and working of a moving coil galvanometer. How can its sensitivity be increased? [3]

b. What is the maximum torque on a galvanometer coil $5 \text{ cm} \times 12 \text{ cm}$ of 600 turns when carrying a current of 10^{-5} A in a field where the magnetic field of 0.1 Tesla is applied? [2]

18. a. State Bohr's postulates for atomic model. [2]

b. Derive the expression for radius of an electron bound to hydrogen atom. [3]

19. a. Explain the Millikan's oil drop experiment to determine the charge of an electron. [3]

b. Specific charge of a particle is $4.4 \times 10^7 \text{ C kg}^{-1}$. It is moving in a circular orbit with velocity $3.52 \times 10^5 \text{ m/s}$ in a magnetic flux density 9.4 T. Find the radius of its orbit. [2]

GROUP-C [3 × 8 = 24]

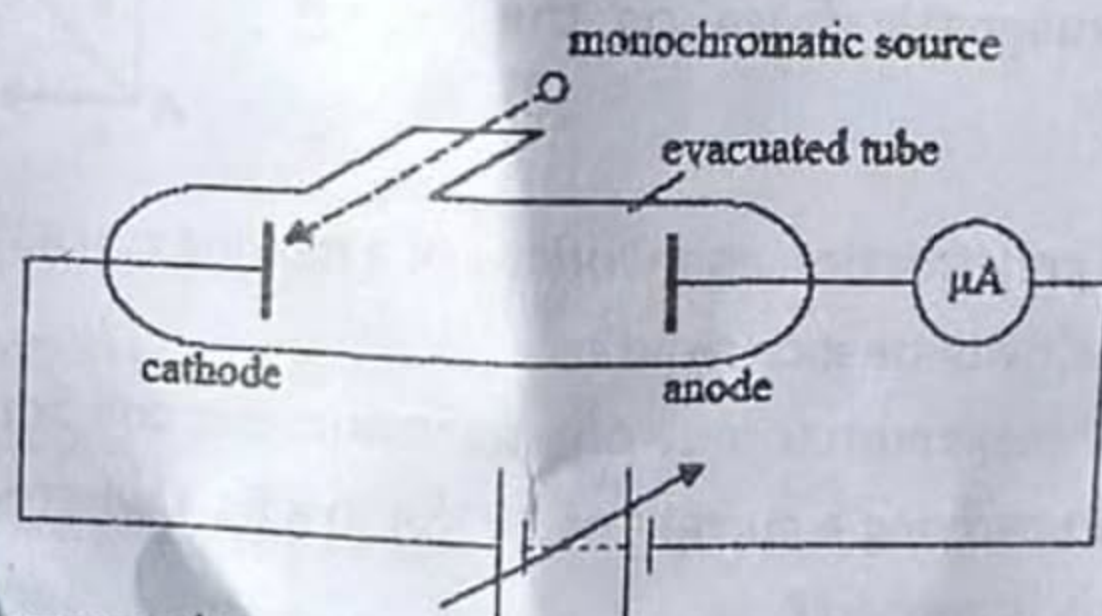
20. a. Define moment of inertia. [1]

b. Derive an expression for the moment of inertia of a thin uniform rigid rod about axis passing through the centre and perpendicular to the length of the rod. [3]

c. State the principle of conservation of angular momentum. [1]

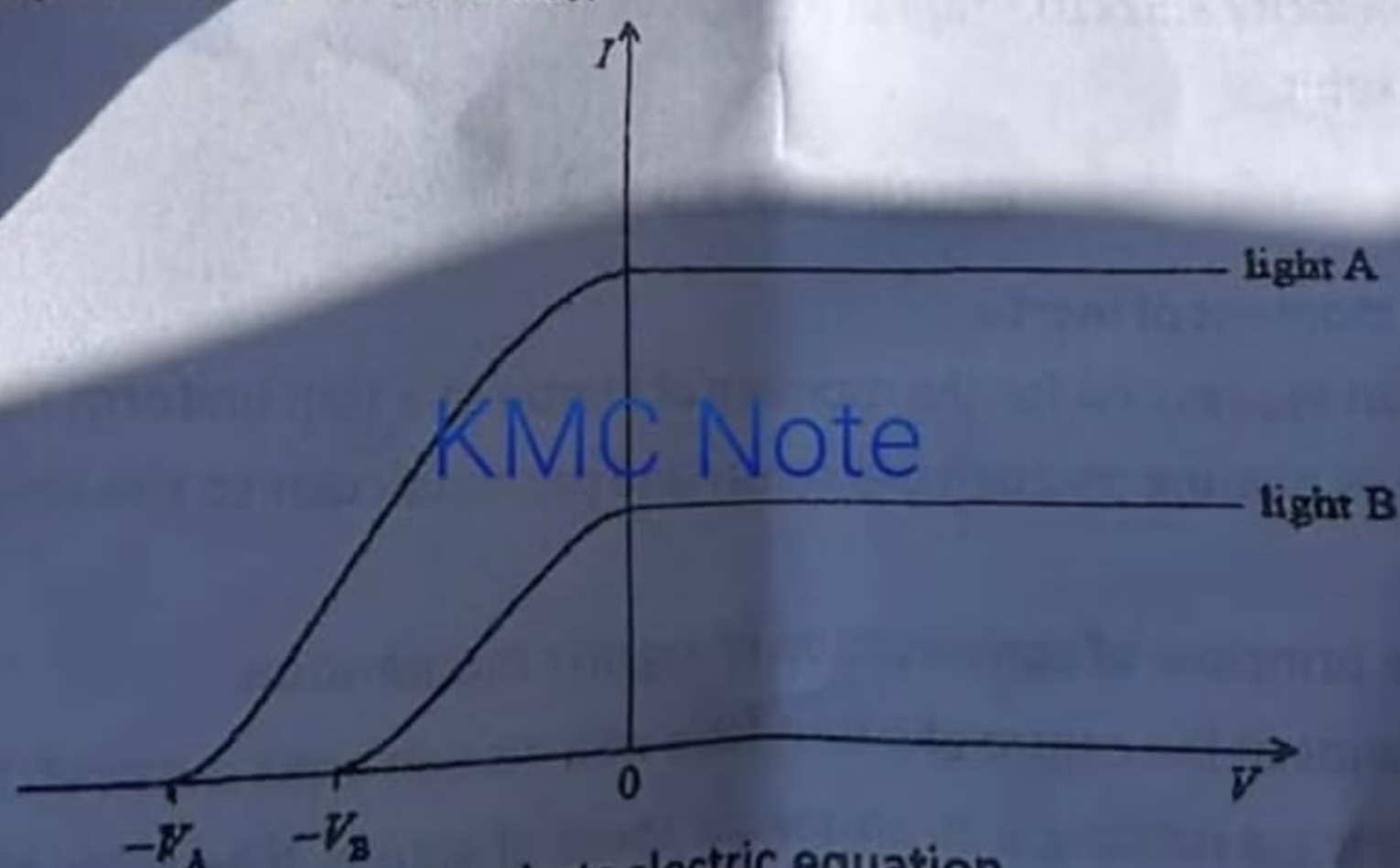
d. A child stands at the centre of a turn table with his two arms outstretched. The turn table is set rotating with an angular speed of 40 rpm. If he folds his hands back and thereby reduces his moments of inertia to $2/5$ times the initial value. Assume that the turn table rotates without friction. Show that the child's new kinetic energy of rotation is more than the initial kinetic energy of rotation. How do you account for this increase in kinetic energy? [3]

21. a. State Biot-Savart law. [2]
 b. Find the magnetic field at a point on the axis of current carrying circular coil. [3]
 c. Two long and parallel straight wires A and B carrying currents of 8.0 A and 5.0 A in the same direction are separated by a distance of 4.0 cm. Estimate the force on a 10 cm section of wire A. [3]
22. The diagram is a representation of apparatus used to study the photoelectric effect.



Light from the monochromatic source is incident on a cathode placed in an evacuated tube.

A variable voltage supply is connected between anode and cathode and the photoelectric current is registered by the microammeter. The sketch graph shows how the photoelectric current I varies with the potential difference V between anode and cathode for two sources of light, A and B, of different frequencies and intensities.



- (a) Explain Einstein's photoelectric equation. [2]
 (b) Explain with reference to the Einstein model, which graph, A or B, corresponds to the light with the greater frequency. [2]
 (c) The frequency of the light that produces graph A is 8.8×10^{14} Hz. The magnitude of V_A is 1.6 V.

- (i) State the value of the maximum energy, in eV, of the electrons emitted from the cathode. [1]
- (ii) Determine the work function, in eV, of the surface of the cathode. [1]
- (d) The frequency of the incident light is increased but the intensity remains constant.
Explain why this increase in frequency results in a change to the maximum photoelectric current (saturation current). [2]

"Best of Luck"

KMC Note

Class: XII 'Science'

Time: 25 Min.

SET-A

Subject: Chemistry (3021)

Tick the best options among the four options. Apply tick on two options will be automatically denied and mark will not be counted.

Symbol no.:

KMC Note

Invigilator's Sign:

Re-write the correct option of each question in your answer sheet

GROUP-A (MCQ) [11×1 = 11]

- The volume of water that must be added to 50 ml of 49 gm/lit sulphuric acid solution to make it exactly semi normal is
a) 100 ml ☒ b) 50 ml c) 150 ml d) 200 ml
- The equivalent mass of H_3PO_4 in the reaction,
 $H_3PO_4 + Ca(OH)_2 \rightarrow CaHPO_4 + 2H_2O$ is
a) 98 b) 32.6 ☒ c) 49 d) 37
- A buffer solution of pH=9 can be prepared by mixing
a) CH_3COONa & CH_3COOH b) $NaCl$ & $NaOH$
☒ c) NH_4Cl & NH_4OH d) KH_2PO_4 & H_3PO_4
- The pH of a neutral solution at $50^\circ C$ is ($pK_w=13.26$)
☒ a) 6.63 b) 6.03 c) 7.13 d) 7.0
- The most stable oxidation state of chromium is
a) +5 ☒ b) +3 c) +2 d) +4
- Formula of Schweitzer's reagent is
☒ a) $[Cu(NH_3)_4]SO_4 \cdot 4H_2O$ b) $[Cu(NH_3)_4]SO_4 \cdot 5H_2O$
c) $[Cu(NH_3)_4]SO_4 \cdot H_2O$ d) $[Cu(NH_3)_4]SO_4$
- Which of the following is used to produce lithopone?
a) $ZnCO_3$ b) ZnO c) $ZnCl_2$ ☒ d) $ZnSO_4$
- Lucas reagent is
☒ a) anhydrous $ZnCl_2$ + Conc. HCl b) conc. H_2SO_4
c) alc. KOH d) $Zn - Hg/HCl$
- Chloral is needed for the synthesis of
a) Benzal Chloride ☒ b) DDT c) Chlorobenzene d) Gammexane
- Acetic anhydride reacts with diethyl ether in the presence of anhydrous $AlCl_3$ to give
a) CH_3COOCH_3 b) $CH_3CH_2COOCH_3$
☒ c) $CH_3COOCH_2CH_3$ d) CH_3CH_2OH
- The electrophile in nitration is
a) nitrate ion b) nitrite ion c) cyanide ion ☒ d) nitronium ion

KMC Note

Time: 25 Min.

Subject: Chemistry (3021)

Time: 25 Min. Subject: Chemistry (3021)
Tick the best options among the four options. Apply tick on two options will be automatically denied and mark will not be counted. **KMC Note**

Symbol no.:

Invigilator's Sign:

Re-write the correct option of each question in your answer sheet.

GROUP-A (MCQ) [11×1 = 11]

- Acetic anhydride reacts with diethyl ether in the presence of anhydrous $AlCl_3$ to give
 - CH_3COOCH_3
 - $CH_3CH_2COOCH_3$
 - $CH_3COOCH_2CH_3$
 - CH_3CH_2OH
- Lucas reagent is
 - anhydrous $ZnCl_2 + Conc. HCl$
 - conc. H_2SO_4
 - alc. KOH
 - $Zn - Hg/HCl$
- Chloral is needed for the synthesis of
 - Benzal Chloride
 - DDT
 - Chlorobenzene
 - Gammexane
- The electrophile in nitration is
 - nitrate ion
 - nitrite ion
 - cyanide ion
 - nitronium ion
- The pH of a neutral solution at $50^\circ C$ is ($pK_w = 13.26$)
 - 6.63
 - 6.03
 - 7.13
 - 7.0
- A buffer solution of pH=9 can be prepared by mixing
 - CH_3COONa & CH_3COOH
 - $NaCl$ & $NaOH$
 - NH_4Cl & NH_4OH
 - KH_2PO_4 & H_3PO_4
- The equivalent mass of H_3PO_4 in the reaction, $H_3PO_4 + Ca(OH)_2 \rightarrow CaHPO_4 + 2H_2O$ is
 - 98
 - 32.6
 - 49
 - 37
- The volume of water that must be added to 50 ml of 49 gm/lit sulphuric acid solution to make it exactly semi normal is
 - 100 ml
 - 50 ml
 - 150 ml
 - 200 ml
- Formula of Schweitzer's reagent is
 - $[Cu(NH_3)_4]SO_4 \cdot 4H_2O$
 - $[Cu(NH_3)_4]SO_4 \cdot 5H_2O$
 - $[Cu(NH_3)_4]SO_4 \cdot H_2O$
 - $[Cu(NH_3)_4]SO_4$
- The most stable oxidation state of chromium is
 - +5
 - +3
 - +2
 - +4
- Which of the following is used to produce lithopone?
 - $ZnCO_3$
 - ZnO
 - $ZnCl_2$
 - $ZnSO_4$

Class: XII 'Science'

F.M: 75

Time: 3hr.

Subject: Chemistry [3021]

P.M: 30

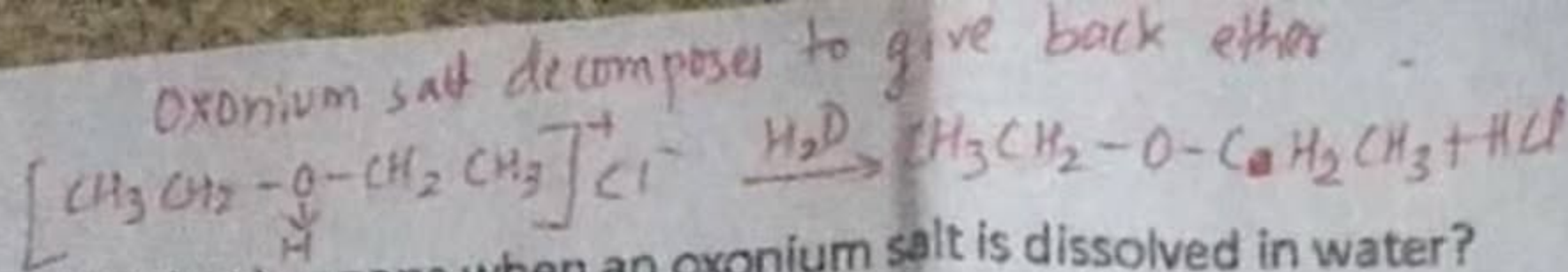
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Attempt all the questions.

GROUP B [8×5=40]

Very short questions

1. a) State law of chemical equivalence and hence derive normality equation. [1+2]
 $V_1 N_1 = V_2 N_2$ or $V_1 S_1 = V_2 S_2$
 b) 7.875 gm hydrated oxalic acid crystals are dissolved in 250 ml water. 16 ml of this solution neutralizes 40 ml of 0.2N NaOH. Calculate the number of water of crystallization. 2 [2]
2. a) Derive an expression of Ostwald's dilution law for weak electrolyte and write its limitation. $\alpha = \sqrt{\frac{K_{eq}}{C}}$ or $\alpha = \sqrt{K_{eq} \times V}$ Applicable only for weak electrolyte [2+1]
 b) The pH of KOH is 8. If it is diluted 1000 times, calculate the pH of diluted base. [2]
 $pOH = 6.995$ $pH = 7.004$
3. a) Discuss the effect of temperature and catalyst in reaction rate. [1+2]
 b) Differentiate between average rate of reaction and instantaneous rate of reaction. At least 3 points. $\rightarrow$ certain interval of time $\rightarrow$ particular instant of time [2]
 $\rightarrow$ Energy profile diagram for catalyzed & uncatalyzed rxn is reqd.
4. a) Write short note on 'Extraction of Zinc from its chief ore.' [4]
 b) What happens when zinc is exposed to moist air? $Zn(OH)_2$ is formed [1]
 Basic zinc carbonate
5. List out the main general characteristics of Transition metals with relevant examples. [5]
6. a) Starting from trichloromethane, how would you prepare
 i) Ethyne ii) Chloropicrin iii) chloretone [1+1+1]
 b) How would you distinguish ethanol from propan-1-ol. [2]
7. a) Write the formula of various positional and functional isomers of alcohols having molecular formula C_3H_8O with IUPAC name. [2]
 b) How would you distinguish these isomers by Victor Meyer's method? [3]
8. a) Give reason
 i) It is dangerous to boil old sample of ether. $CH_3CH_2OCH_2CH_3 + O_2 \xrightarrow{\text{Sunlight}} CH_3CH_2-O-O-CH_2CH_3$ peroxide is formed [1+1]
 ii) Ether is stored in bottle containing iron wire.
 Iron has greater affinity towards oxygen than ether has.
 So, iron combines with oxygen and formation of peroxide is controlled.



- b) What happens when an oxonium salt is dissolved in water? [1]
- c) Why chlorobenzene is less reactive as compared to benzene towards electrophilic substitution reaction? [2]

KMC Note

Group C [3x8=24]

Long Questions

9. a) What is the role of dil. HCl in the qualitative analysis of Group-II Metal sulphide? [3]

Explain. In the selective precipitation of Gr II Metal sulphide. Explanation reqd.

b) 200 ml of $1.3 \times 10^{-3} M AgNO_3$ is mixed with 100 ml of $4.5 \times 10^{-5} M Na_2S$. Will precipitation occur? (K_{sp} of $Ag_2S = 1.6 \times 10^{-49}$) Yes ppt occurs. [3]

c) Explain the concept of relative strength of acid and base with relevant examples on the basis of Bronsted-Lowry concept. Strong bronsted acid will form weak conjugate base & vice versa. [2]

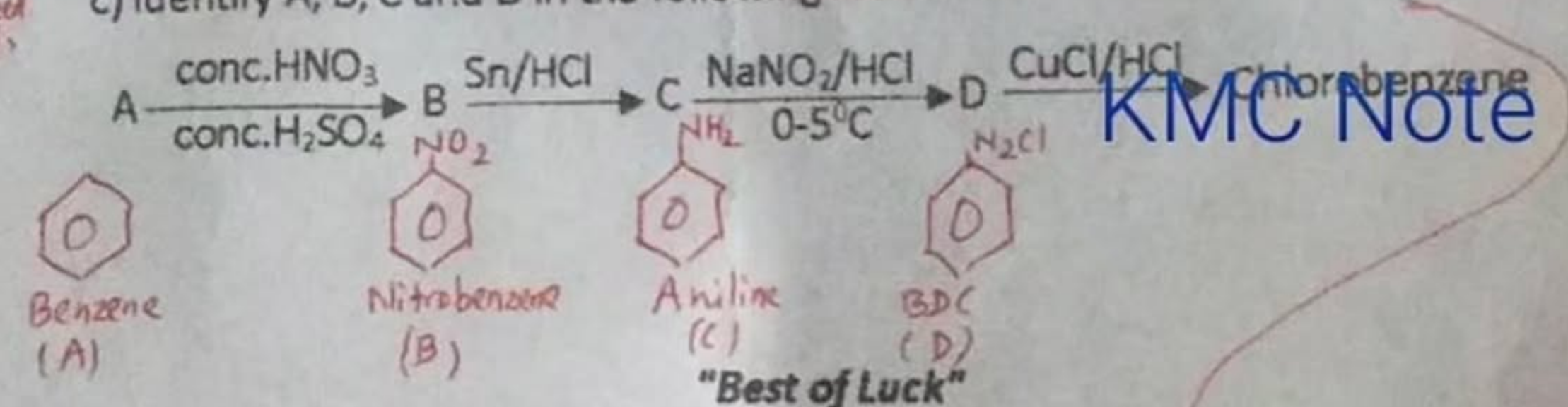
10. a) Write short note on "Chemistry of Blue Vitriol". [5]

b) What is 'matte'? How is it formed? What is its function? [3]

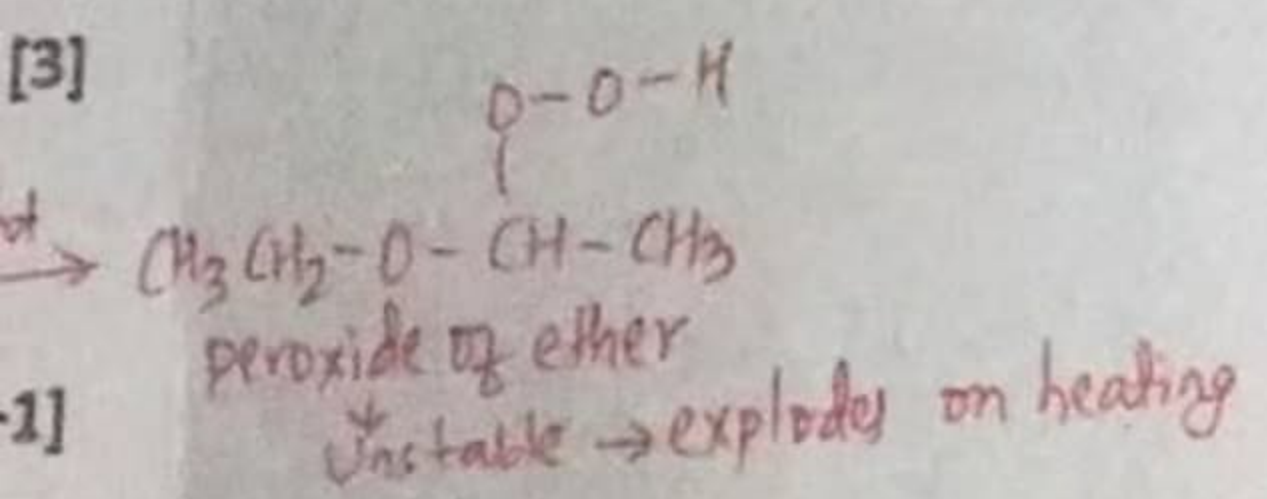
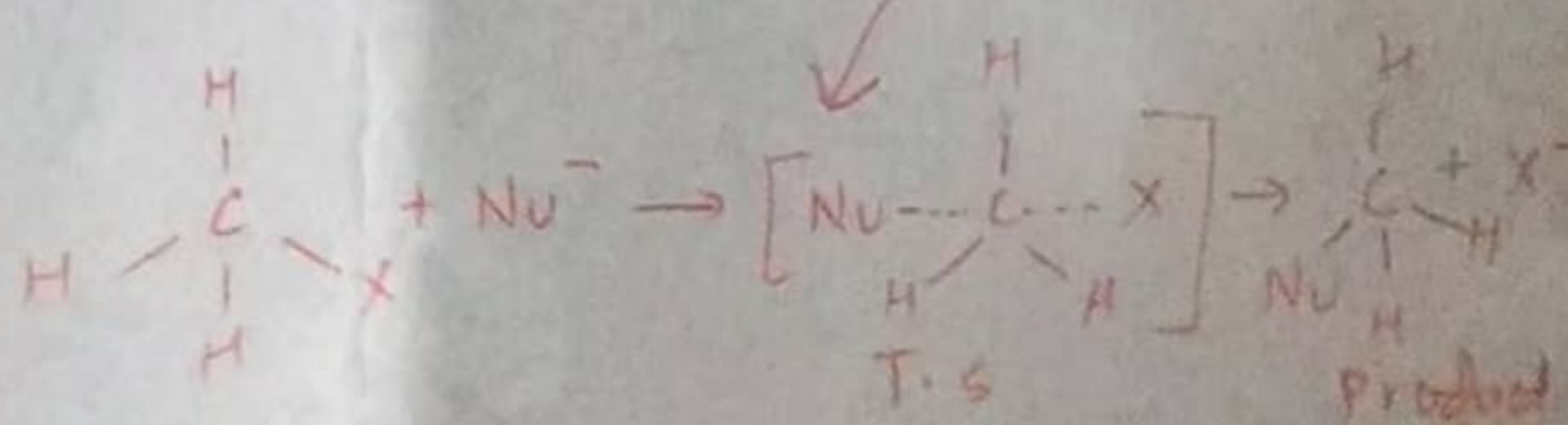
11. a) Write short note on "Reduction of nitrobenzene in various medium" [4]

b) Explain SN_2 reaction. (i) Rate $\propto [R-X][Nu^-]$ (ii) 2nd order (iii) Reactivity order: $CH_3X > 1^\circ > 2^\circ > 3^\circ$ alkyl halide [2]

c) Identify A, B, C and D in the following reaction. [2]



KMC Note





KATHMANDU MODEL SECONDARY SCHOOL

FIRST TERM EXAMINATION 2081

Class: XII 'Science'

SET-A

Time: 25 mins.

Subject: Com. Mathematics (0081)

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

GROUP-A (MCQ) [11×1 = 11]

Re-write the correct option of each question in your answer sheet

1. What is the value of $(\sin\theta + i \cos\theta)^n$?

A) $\sin\theta + i \cos\theta$

B) $\cos\theta + i \sin\theta$

C) $\cos\left(\frac{n\pi}{2} - \theta\right) + i \sin\left(\frac{n\pi}{2} - \theta\right)$

D) $\cos\left(\frac{n\pi}{2} - n\theta\right) + i \sin\left(\frac{n\pi}{2} - n\theta\right)$

2. Which one of the following system of linear equation is consistent and independent?

A) $3x + 4y = 5$

B) $x + y = 5$

C) $3x + 4y = 10$

D) $4x + 4y = 20$

C) $3x + 4y = 7$

D) $3x + 4y = 7$

C) $3x + 4y = 5$

D) $x + y = 5$

3. In Gauss-seidal method for solving

KMC Note

$3x + y = 5$ and $x + 5y = 11$, $(x_0, y_0) = (0, 0)$, what is (x_1, y_1) ?

A) (5,11)

B) (11,5)

C) $\left(\frac{5}{3}, \frac{11}{5}\right)$

D) $\left(\frac{3}{5}, \frac{5}{11}\right)$

4. If $\frac{(n+1)!}{(n-2)!} = 720$, what is the value of n ?

A) 8

B) 9

C) 10

D) 11

5. If $C(n, 10) = C(n, 15)$, what is the value of $C(n, 25)$?

A) 10

B) 25

C) 1

D) 5

6. In the expansion of $(4 - 4x + x^2)^{20}$, how many number of terms?

A) 40

B) 21

C) 41

D) 40

7. What is the value of $\vec{i}(\vec{j} \times \vec{k}) + \vec{j}(\vec{k} \times \vec{i}) + \vec{k}(\vec{i} \times \vec{j})$?

A) 1

B) 2

C) 0

D) 3

8. Under what condition a line $cx + dy = k$ will tangent to the circle

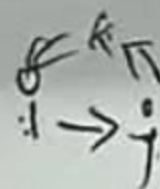
$x^2 + y^2 = a^2$?

A) $k = \pm \sqrt{c^2 + d^2}$

B) $k = \pm a \sqrt{c^2 + d^2}$

C) $k = a \pm \sqrt{c^2 + d^2}$

D) $k^2 = \pm a \sqrt{c^2 + d^2}$



the equation of the normal to the curve $y = \sin x$ at $(0,0)$ is

A) $x = 0$

B) $y = 0$

C) $x + y = 0$

D) $x - y = 0$

If the rate of change of area of a circle is equal to the rate of change of its diameter, then its radius is equal to

A) $\frac{2}{\pi}$ unit

B) $\frac{1}{\pi}$ unit

C) $\frac{\pi}{2}$ unit

D) π units

1. What is the value of $\int \frac{1}{2x^2+25} dx$

A) $\frac{1}{5} \tan^{-1} \left(\sqrt{\frac{2x}{5}} \right) + c$

B) $\frac{1}{5\sqrt{2}} \tan^{-1} \left(\sqrt{\frac{2x}{5}} \right) + c$

C) $\frac{1}{\sqrt{2}} \tan^{-1} \left(\sqrt{\frac{2}{5}} x \right) + c$

D) $\frac{1}{5\sqrt{2}} \tan^{-1} \left(\frac{\sqrt{2}}{5} x \right) + c$

KMC Note

Best of Luck

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Knowledge Through Quality Education
KATHMANDU MODEL SECONDARY SCHOOL
 FIRST TERM EXAMINATION 2081

Class: XII 'Science'

SET-B

Time: 25 mins.

Subject: Com. Mathematics (0081)

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Sh
12

GROUP-A (MCQ) [11×1 = 11]

Re-write the correct option of each question in your answer sheet.

1. If $\frac{(n+1)!}{(n-2)!} = 720$, what is the value of n ?
 A) 11 B) 10 C) 9 D) 8
2. If $C(n, 10) = C(n, 15)$, what is the value of $C(n, 25)$?
 A) 1 B) 5 C) 10 D) 25
3. In Gauss-seidal method for solving
 $3x + y = 5$ and $x + 5y = 11$, $(x_0, y_0) = (0, 0)$, what is (x_1, y_1) ?
 A) $(\frac{3}{5}, \frac{5}{11})$ B) (11,5)S
 C) $(\frac{5}{3}, \frac{11}{5})$ D) (5,11)
4. It is given that $z = r(\cos\theta + i \sin\theta)$ and n be a positive integer. Which one of the following represent z^n ? KMC Note
 A) $r[\cos n\theta - i \sin n\theta]$ C) $r^n[\cos n\theta - i \sin n\theta]$
 B) $r^n[\cos n\theta + i \sin n\theta]$ D) $r[\cos n\theta + i \sin n\theta]$
5. Which one of the following system of linear equation is consistent and independent?
 A) $3x + 4y = 7$ B) $x + y = 5$
 $x + y = 5$ C) $4x + 4y = 20$
 C) $3x + 4y = 7$ D) $3x + 4y = 5$
 $3x + 4y = 5$ D) $3x + 4y = 10$
6. In the expansion of $(4 - 4x + x^2)^{20}$, how many number of middle term(s)?
 A) 40 B) 1 C) 41 D) 2
7. If the rate of change of area of a circle is equal to the rate of change of its diameter, the its radius is equal to
 A) $\frac{\pi}{2}$ unit B) $\frac{1}{\pi}$ unit C) $\frac{2}{\pi}$ unit D) π units

8. Which of the following is an integral value of $\int \frac{dx}{4x^2+1}$?
- A) $2 \tan^{-1}\left(\frac{x}{2}\right) + c$ B) $2 \tan^{-1}(2x) + c$
C) $\tan^{-1}\left(\frac{x}{2}\right) + c$ D) $\frac{1}{2} \tan^{-1}(2x) + c$
9. The equation of the normal to the curve $y = \sin x$ at $(0,0)$ is
- A) $x = 0$ B) $y = 0$ C) $x + y = 0$ D) $x - y = 0$
10. It is given that $\vec{a}$ and $\vec{b}$ are non-zero vectors and $\frac{\theta}{2}$ be the angle between $\vec{a}$ and $\vec{b}$. Which one of the following represents $\vec{a} \cdot \vec{b}$?
- a) $ab \cos \theta$ B) $ab \sin \theta$ C) $ab \cos \frac{\theta}{2}$ D) $ab \sin \frac{\theta}{2}$
11. Under what condition a line $cx + dy = k$ will be tangent to the circle $x^2 + y^2 = a^2$?
- A) $k^2 = \pm a\sqrt{c^2 + d^2}$ B) $k = \pm a\sqrt{c^2 + d^2}$
C) $k = a \pm \sqrt{c^2 + d^2}$ D) $k = \pm \sqrt{c^2 + d^2}$

Best of Luck

Academic Excellence Through Quality Education
KATHMANDU MODEL SECONDARY SCHOOL
FIRST TERM EXAMINATION 2081

Class: XII 'Science'

Time: 3hr.

Subject: Mathematics (0081)

F.M: 75

P.M: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

GROUP-B [8×5 = 40]

Short Answer Questions.

12. a) What is the value of $c(n, r) + c(n, r-1)$? [1]
 b) Write one application of De-moivre's theorem with an example. [2]
 c) A delegation of 4 students is to be selected from a total of 12 students.
 In how many ways can the delegation be selected
 i) if all the students are equally willing? [1]
 ii) if two particular students have to be in the delegation? [1]
13. Find the number of ways in which 5 girls and 5 boys can be arranged in a row
 a) if no two boys are together. [1]
 b) if boys and girls are alternate. [2]
 c) if all the girls sit together and all the boys sit together. [2]
14. In the expansion of $\left(2x^2 - \frac{1}{3x}\right)^{15}$,
 a) write the number of terms. [1]
 b) what are binomial coefficients? [1]
 c) find the general term. [1]
 d) find the coefficient of x^9 (if any). [1]
 e) find the term free from x (if any). [1]
15. a) If $\vec{a} = 3\vec{i} + 4\vec{j} - 5\vec{k}$ and $\vec{b} = 5\vec{i} - 6\vec{j} + \vec{k}$ are two vectors, find $\vec{a} \times \vec{b}$ and show that it is perpendicular to both $\vec{a}$ and $\vec{b}$. [2]
 b) The dot product of two vectors and cross product of two vectors are interrelated. Justify the statement. [3]
16. By using simplex method, find the optimal value and optimal solution of [5]
 Max. $z = 5x - 3y$
 Subject to $3x + 2y \leq 6$
 $-x + 3y \geq -4$
 $x, y \geq 0$
17. a) Write $\tanh x$ in terms of exponential function. [1]
 b) If $y = 2 \tan^{-1}\left(\tanh \frac{x}{2}\right)$, show that $\frac{dy}{dx} = \operatorname{sech} x$ [2]
 c) Write one application of derivative to evaluate limits with an example. [2]
18. a) If $\int f(x)dx = g(x) + c$ then express it from derivative point of view. [1]

KMC Note

- b) Evaluate $\int \frac{1}{2x^2-9} dx$ c) Evaluate $\int \frac{7x-9}{x^2-2x+35} dx$ [1+3]
 19. a) If the rate of change of volume of sphere is equal to the rate of change of its radius, find the radius. [2]
 b) If the area of an expanding circular region increases at a constant rate with respect to time then show that the rate of increase of the perimeter with respect to time varies inversely as radius. [3]

GROUP-B [3×8 = 24]

Long Answer Questions.

20. a) Find the value of k for which the system of linear equations have no solution. [3]
 $kx - 2y + z = 1, x - 2ky + z = -2, x - 2y + kz = 1$
 b) A mixture (in kg) is to be made from three foods A, B and C which contain nutrients P, Q and R as shown in the table below. Find the amount of each food which should be mixed in the final output. The available units of P, Q and R are 45, 54 and 45.

Foods	Units of nutrients per kg of the food		
	P	Q	R
A	2	2	4
B	3	5	0
C	4	3	5

- If the cost per kg of the food A, B and C are Rs.40, Rs.60 and Rs. 80 respectively, Find by matrix method the total cost of the mixture. [5]
 21. a) Show that the middle term in the expansion of

$$\left(x + \frac{1}{2x}\right)^{2n+2} \text{ is } \frac{1.3.5 \dots (2n+1)}{(n+1)!}$$

KMC Note

Also, by using it verify that the middle term of

$$\left(x + \frac{1}{2x}\right)^{500} \text{ is } \frac{1.3.5 \dots 499}{(250)!}$$

- [3+1]
 b) Find the condition for the line $y = mx + c$ to be tangent to the circle $x^2 + y^2 = a^2$. Also, write the equation of tangent in slope form. [3+1]
 22. a) Prove that $(\vec{a} - \vec{b}) \times (\vec{a} + \vec{b}) = 2(\vec{a} \times \vec{b})$ and interpret it geometrically. [2]
 b) Find the equations of tangents to the circle $x^2 + y^2 = 25$ which passes through the point (7,1). [2]
 c) Find the angle at which the curves $y = \sin x$ and $y = \cos x$ intersect in $[0, \pi]$. [2]
 d) Show that the curves $x^3 - 3xy^2 = a$ and $3x^2y - y^3 = b$, where a and b are constants, cut each other at right angle. [2]

$\frac{dn^3}{dn} = 3 \frac{dny}{dn}$

Best of Luck

प्रथम त्रैमासिक परीक्षा - २०८१

कक्षा : १२ (विज्ञान संकाय)

समय : ३ घण्टा

पूर्णाङ्क : ७५

उत्तीर्णाङ्क : ३०

विषय : सामाजिक अध्ययन तथा जीवनोपयोगी शिक्षा [Sol. 0061]

विद्यार्थीहरूले सकेसम्म आफ्नै शब्दमा उत्तर दिनुपर्ने छ । दायाँ किनारामा दिइएको अङ्कले पूर्णाङ्क जनाउँदछ ।

सबै प्रश्नहरूको उत्तर दिनुहोस् ।

समूह-क (११×१=११)

तलका प्रश्नहरूको अति छोटो उत्तर दिनुहोस् :

१. आस्तिक दर्शनको मूल आशय के हो ?
२. मानसिक रूपले स्वस्थ रहन तपाईंले गर्दै आउनुभएका कुनै दुई अभ्यास लेख्नुहोस् ।
३. शुन्य सहसम्बन्ध भन्नाले के बुझिन्छ ?
४. प्रतिलिपि अधिकारलाई एक वाक्यमा परिभाषित गर्नुहोस् ।
५. आत्मालोचनाले अन्तर्वैयक्तिक सम्बन्ध सुधारमा कसरी मद्दत गर्दछ ? एक वाक्यमा उल्लेख गर्नुहोस् ।
६. किशोरावस्थालाई किन आँधी-तुफानको उमेर भनिन्छ ? एक वाक्यमा आफ्नो धारणा व्यक्त गर्नुहोस् ।
७. 'गार्हपत्य' र 'आहवनीय' भन्नाले के-के बुझिन्छ ?
८. सूचना समाजलाई एक वाक्यमा परिभाषित गर्नुहोस् ।
९. सुदुरपश्चिम प्रदेशका कुनै दुई सांस्कृतिक विशेषताहरू उल्लेख गर्नुहोस् ।
१०. सम्भावित विपत्तिबाट बच्न तपाईंले पूर्वतयारी कसरी गर्नुहुन्छ ? एक वाक्यमा लेख्नुहोस् ।
११. लिच्छविकालमा निर्मित कुनै दुई वास्तुकलाको नाम लेख्नुहोस्, जो आज इतिहासको पानामा मात्र सीमित छन् ।

समूह-ख (८×५=४०)

तलका प्रश्नहरूको छोटो उत्तर दिनुहोस् :

१२. पूर्वीय चिन्तन परम्पराले आधुनिक संसारलाई पुर्‍याएको योगदान उल्लेख गर्दै दुई साथीका विचमा भएको एउटा संवाद तयार गर्नुहोस् ।

वा

कार्यकारण सम्बन्ध र सहसम्बन्धको महत्त्व उल्लेख गर्दै एउटा संवाद तयार गर्नुहोस् ।

१३. शाब्दिक प्रस्तुतीकरणलाई प्रभावकारी बनाउने कुनै पाँच उपायहरू उल्लेख गर्नुहोस् ।

१४. समानुभूति सिप विकास गर्न अवलम्बन गर्न सकिने उपायका सन्दर्भमा आफ्नो साथीलाई सुझाव प्रस्तुत गर्नुहोस् ।

१५. आत्मबोध र आत्ममूल्याङ्कनले व्यक्तित्व विकासमा कसरी सहयोग गर्दछ ? व्याख्या गर्नुहोस् ।

१६. "सामाजिक सम्बन्धले सामाजिक अन्तरनिर्भरता स्थापित गर्न महत्त्वपूर्ण भूमिका खेल्दछ ।" यो भनाइलाई उदाहरणसहित प्रष्ट पार्नुहोस् ।

१७. गण्डकी प्रदेशको आर्थिक कृषिकलापमा पर्यटन उद्योगको कस्तो सम्भावना छ ? प्रष्ट पार्नुहोस् ।

१८. तपाईं स्थानीय तहको प्रमुख हुनुभयो भने विपत्तिको व्यवस्थापन कसरी गर्नुहुन्छ ?

१९. किरातकालिन सामाजिक तथा आर्थिक अवस्थालाई बुँदागतरूपमा प्रस्तुत गर्नुहोस् ।

वा

लिच्छविकालीन इतिहासबाट हालका पुस्ताले के-के कुरा सिक्न सक्नेछन् ? आफ्नो जानकारीलाई सूचीबद्ध गर्नुहोस् ।

समूह-ग (३×८=२४)

तलका प्रश्नको लामो उत्तर दिनुहोस्:

२०. कुनै एक भौगोलिक क्षेत्रको अध्ययन भ्रमण गर्नका लागि प्रस्तावनाको नमूना तयार पार्नुहोस् ।

२१. "सिर्जनात्मक चिन्तन सिप : विश्वमा सबैभन्दा बढी माग हुने सिप" शीर्षकमा राष्ट्रिय दैनिक पत्रिकाको लागि सम्पादकीयको नमूना तयार पार्नुहोस् ।

वा

तपाईंले आफ्नो मार्गनिर्देशक दर्शनको रूपमा कुन दर्शन अवलम्बन गर्नु भएको छ ? उक्त दर्शनले तपाईंको जीवनमा पारेका प्रभावहरू उदाहरणसहित व्याख्या गर्नुहोस् ।

२२. लुम्बिनी प्रदेशको भौगोलिक, आर्थिक र सामाजिक अवस्थाको चित्रण गर्नुहोस् ।

❀ समाप्त ❀